Data Rescue for Historic Scientific Literature: Automatically Extracting and Cross-Validating Published Records

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***Abstract*—Vision-language models afford new opportunities for unlocking historical datasets that were previously locked in unreadable formats, such as scanned PDFs. Low-resource text extractors also benefit communities and researchers interested in data sovereignty, IP and data restrictions, and rising cost concerns. Here, we present a comparative evaluation of 13 different VLM, OCR, and text-based extractors on a long-term ecological dataset with divergent data provenance, run entirely on a local GPU grid. We also explore multiple validation techniques for tabular data, including manual validation, multi-model output and authoritative resource cross-checking. Our results show that Chandra, a multi-model VLM based on Marker and Qwen3.5, outperforms all other extractors, but that programmatic cross-checking and non-VLM input by deterministic OCR is necessary for validation and cleaning. We conclude with identified errors in the published record, a pointer to discordance between the two versions of the dataset, and a full evaluation sweep of the extractors.**

***Index Terms*—OCR, VLM, table extraction, text extraction, pipeline, data, validation, ornithology, ecology**

I. Introduction

Poor data management is one of the failure modes for long-term studies, and may affect use, re-use, and efficacy of the data [1]. Studies have shown that the transfer from paper to digital records can lead to loss of data, generally with an error rate of around 4% [2]. Data cleaning is expensive but necessary, although not all types of data can be easily cleaned [3].

Mitigation options include recollecting the data (unrealistic in many cases), double entry or multiple entry (expensive) [4]–[7] and crowd-sourcing transcriptions (expensive and labour intensive) [8], [9]. The ultimate question we are addressing in this paper is: how do we rescue such data from datasets that have changed over time and accumulated errors and differing data treatments?

In natural sciences, automated pipelines have largely been used in museum contexts for working with natural history specimens and label data [9]–[13]. However, little work has focused on historical tabular data using modern OCR extractors. LLMs present an opportunity for the field [14], but risks include legal and security uncertainty with using externally run LLMs, including API usage through Amazon AWS as in [12], [13]. An alternative process would be to use open source and local VLMs, such as the Qwen family of models [15]–[17] released by Alibaba. The authors of [10] used Donut-base [18], a pre-trained visual document understanding (VDU) transformer, to extract data labels from historical specimens. Other work on historical documents includes [19]–[22]. However, to date no work that we are aware of focuses on extracting natural history data preserved in tabular data formats using VLMs or modern OCR techniques.

The historical data archive of the New Zealand Beach Patrol Scheme (BPS) serves as a long term, high-variance case study to validate this work. The Beach Patrol Scheme is one of the longest-running biological surveys in New Zealand which utilizes the long-term efforts of community scientists. BPS involves volunteers across the country walking beaches and reporting a) the distance of their walks and b) the number, species, and condition of dead birds found. This is useful for tracking ”wrecks”, which are large strandings of dead birds that may occur to different species in different years, and for setting a baseline for the number of dead birds seen by a population.

Data were recorded on paper (Figure 1) that was mailed to a centralized curator. Starting from 1983, these cards were manually digitized into a database [23], with the dataset becoming digital-first in the last two decades. Submissions are now gathered through an online form. The data are relatively large for an ecological dataset, at 100k rows.

A second view of the BPS dataset from the same source data exists: the annual summary reports published in the *Notornis* [24] journal. These summaries were manually compiled from

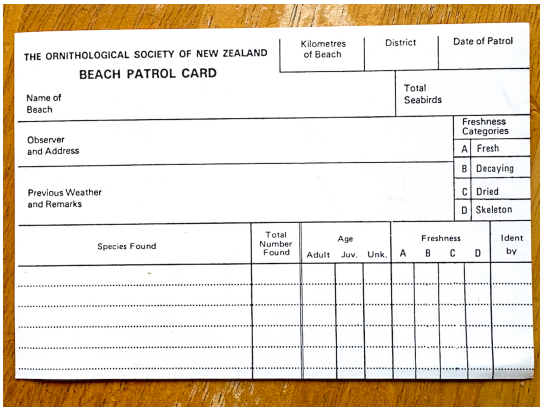

THE ORNITHOLOGICAL SOCIETY OF NEW ZEALAND
BEACH PATROL CARD
Kilometres of Beach
District
Date of Patrol
Name of Beach
Total Seabirds
Observer and Address
Freshness Categories
A Fresh
B Decaying
C Dried
D Skeleton
Previous Weather and Remarks
Species Found
Total Number Found
Age
Adult Juv. Unk.
Freshness
A B C D
Ident by

Fig. 1. An example BPS card.

the original card datasets, but the process was opaque and may have changed over the years. This process resulted in differences in errors in transcription. Errors may have come from calculation or formatting of summaries during the reporting process, and also from the digitization of the cards by hand. The two datasets are not in agreement on all numbers.

In order to determine the level of disagreement, the summary tables need to be digitized. Doing this by hand for 40 years of publications would be an arduous process. Using an external LLM such as [25], [26] is problematic for this research for three reasons:

- **Indigenous and Data Sovereignty**: As pointed out by [12], LLM usage in this context is risky for reasons of legality and Indigenous sovereignty. All bird species in Aotearoa New Zealand are taonga 'sacred' in te ao Māori, the world of the Indigenous people of New Zealand [27]. Practitioners working under the perspective of kaitiakitanga 'guardianship' should consider the implications of the downstream use of the data, and refrain from participating in extractive technologies [28]–[31].
- **Copyright**: The data are copyrighted by the Ornithological Society of New Zealand; although the PDFs are publicly available on the website, a `robots.txt` file on the BPS website [32] indicates a desire for papers to not be scraped automatically.
- **Cost**: Finally, LLMs are resource-intensive tools that may not be necessary for localized extraction of text. Smaller, targeted models may have better application for some tasks than generalized tools [11].

In this paper we used an OCR pipeline and a local GPU cluster to automatically extract the data from the published articles of the journal while remaining compliant with the three concerns noted above. We used both the older scanned PDFs of the journals and newly re-scanned images of a subset of the journal articles.

We used 13 different text extractors on the PDFs in a comparative evaluation to find the best candidate for a full extraction pipeline. As well as resulting in digitized tabular data, this process also results in secondary and tertiary datasets: a list of all species names used in the reports over time, which change as taxonomic understanding shifts, and a complete text output from the summary articles.

We present a pipeline for verifying data using several of these methods in sequence: manual transcription of a ground-truth, a comparative evaluation including multiple types of local-only OCR techniques, automatic tests to verify column and row data, and verification of scientific names against a known dataset. We present a workflow for transcribing, structuring, cleaning, normalizing, verifying, and archiving the data, following best practices in computational methods.

II. Background

Many species are at risk of going extinct, due to anthropogenic factors, habitat loss, and climate change. Long-term studies mapping historical distributions are necessary to understand the current status of these species, for present-day ecological modelling, and to learn how species respond to specific events.

An example is pelagic birds in New Zealand waters - albatrosses, shearwaters, and others - which were historically known to be present across the region in massive numbers. Today, there are fewer seabirds; however, without long-term studies affording insights into changing demographic trends, it is difficult to quantify the loss.

One of the objectives of BPS is "to provide information on the species, location and numbers of all birds found dead on the New Zealand coastline, including islands," and "to record, where possible the reasons, for seabird mortality" [32]. The scheme was formalized in 1951 [23], but records stretch back to 1943. As of January, 2025, there are 106,733 records in the raw BPS database, with a total of 4,337,921 individual birds counted and 1,290,634.1 km of patrolled beaches [33].

The number of beaches patrolled in the database was minimal before 1960: see Fig. 3. From 1960, BPS participants were given physical cards to record their patrols [34], as shown in Fig. 1. From 1960-1999, annual summaries were published in the journal *Notornis* (see Figure 2).

For example, the 1984 report [35] included the tables: (1) Numbers of dead seabirds recovered and kilometers covered on each coast in 1984; (2) Seabirds of which 1 to 15 specimens were found in 1984; (3) Coastal distribution of the seabirds more commonly found dead in 1984; (4) Monthly distribution of the seabirds more commonly found dead in 1984; and a final table and two figures on the recovery rate for Southern Fulmar *Fulmarus glacialoides*. This format is generally followed for other reports, although the formatting of the tables is heterogeneous over time. The tradition of making summaries was discontinued from 2000 until 2007; after this, short reports were published annually on the website, but without the same level of granularity.

The BPS database is currently stored as two separate spreadsheets; one Excel file for records until 2018, and another CSV file sourced from an automated-web system from 2019 (with some recently-added data from 2012 onwards). There is some minor overlap between the data sources. The data are

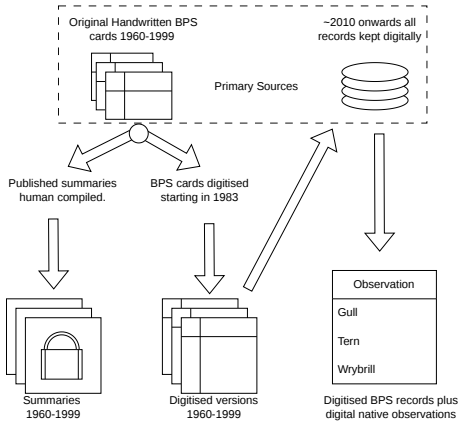


Fig. 2. BPS data records and digitization. After 1983, source records could be either on BPS card or digital.

originally in two databases due to different approaches made by two separate development companies (Ian Armitage (BPS convenor), pers. comm.).

The digitized databases do not conform exactly to the regional totals originally published in annual reviews. For example, in Table 1 from 1990 [36], three of 15 regions show >25% density disagreements for birds found per kilometer of beach: Southland is 2.1x the value in the database; Bay of Plenty 1.5x, and the North Coast of the South Island 1.86x. There are several possible reasons for this discrepancy: errors introduced in transcription, reports which were only filed after the survey was made, errors in the original summation of data, and so on. These errors affect the long-term use of the dataset, and point to wider issues that there are effectively two different datasets: the transcribed dataset from the original physical cards of observations (hereafter **D1**), and the tables of annual summaries from the published journal articles in *Notornis* [24] (**D2**).

Not all records of beached birds in D2 are in D1. For instance, [37] includes a note of a beach walk with a date, length, and list of dead birds, but it is not included in D1. It is also unclear if the count lists all species found during the stranding. Besides data gaps, there are other concerns. For instance, D2 uses miles in 1960-64, 1968-9, and 1973, but kilometers in 1965-7, 1970-2, and 1974 onwards. NZ switched to the metric system gradually from 1969-1976 [38], and it is unclear why the transition was interleaved.

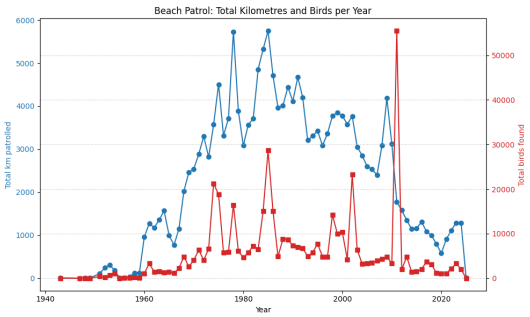


Fig. 3. Total number of birds and beaches patrolled per year

III. Methods

A. Data gathering

The BPS convenor provided D1. The data are available from BirdsNZ (the Ornithological Society of New Zealand) under a Creative Commons Attribution 3.0 New Zealand license [39]. A data agreement waiver provided by the Society has more restrictive terms.

For D2, PDFs of paper summaries from *Notornis* [24] were downloaded from the journal website. Finally, in order to verify species names in Section III-E, the 2022 BirdsNZ Checklist PDF [40] was also downloaded from the same website.

B. Ground Truth

All following code was written using Python [41] or using Bash [42]. The process for creating the ground truth (GT) dataset can be seen in Fig. 4.

In order to judge the accuracy and precision of extractors, the GT with verified tables needs to be created as a control. Tables from the years 1960 [43], 1963 [44], 1974 [45], 1984 [35], 1990 [36], and 1997 [46] were selected for the ground truth dataset. The 1963, 1974, 1984, and 1990 tables [35], [36], [44], [45] were selected for their temporal spread, not for intrinsic qualities; the tables in 1960 and 1997 [43], [46] were specifically selected as the earliest and latest in the series.

Before running the extractors, each paper was assessed for whether the tables were typeset or written using a typewriter or a monospace font. The script ran pdfplumber [47] and checked whether the page source was phototypeset or digitally typeset, and whether there was an extractable text-layer. Only [46]

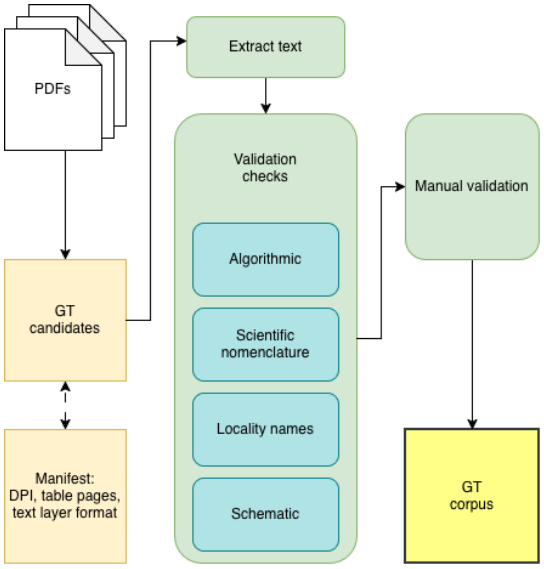


Fig. 4. The process for creating the GT corpus (n=21 tables).

was typeset digitally and rendered as vector text from the beginning, and is machine-readable from the outset. Further, the tables in 1960 and 1963 [43], [44] were both originally typeset, while the remaining tables [35], [36], [45] appear to have been copied from a typewriter. The PDFs contained a text overlay from Adobe Acrobat 6.0 [48], excepting 1997-9, whose vectorized text obviated the need for a scan.

D2 was originally printed and published in magazine format; as such, each article is a PDF of a scanned physical magazine. The scanning quality may also have an impact on the viability of OCR. Because of this, a script determined the dots-per-inch (dpi) value of the page for each table: all tables were scanned at 300dpi, except for 1984 table 1 and 2, and each of the 1990 tables, which varied from 207-315dpi. As these were variable, all tables were also rescanned from physical journals at 400dpi.

Each table for the GT was either manually copied from D2, or Qwen3-VL-8B [17] was used to automatically parse the table. Where this failed, a screenshot was taken and the screenshot was uploaded to Claude.ai Opus-4.8 [49], with the following prompt: "Change nothing. Do not correct any spelling errors or arithmetic errors. Convert this into a CSV for download." The sample size ($<$10 tables) was judged to be small enough to fit under Fair Dealing in NZ copyright law [50].

The output was then loaded into Google Sheets [51] and visually inspected. To match the source files, casing was normalized to match the source; scripts were run to convert blank tables to dashes where needed; and headers were correctly formatted. Where a table had sum rows or columns on the bottom or right side of a table, the entire row or table was summed and compared to the extracted value. This process alone caught errors in the originally printed tables; these errors were logged, and are described in Section IV-B. The necessary rotation for each page was also noted, as some extractors (e.g., olmOCR) may not be able to recover text from a poorly-rotated page. Finally, an index CSV was created for use by the extractors, with columns: pdf_path, pdf_page, rotation, schema_type, typography, text_layer_status, source_dpi, header_row, and verified. Each table was classified into four schema families: A-bimonthly, A-monthly, and A-covered effort tables (the last pairing kilometres and bird counts per region), and B-species lists — recorded in the schema_type column. The schema is used for a layout-specific verifier which checks row, column, and total identities, as referenced in Section IV-B.

Finally, a second GT_2 dataset was made, manually counting the number of pages in the PDFs which contained tables or line-figures, compared against the total number of pages, broken down by year in CSV format. This is used to compare table coverage, as in Table II.

C. Extractors

We ran 13 different OCR extraction methods on tables present in the GT (see Figure 5). The methods were chosen to cover a wide range of different types of extractors. Each of these needs to adequately implement OCR, text extraction and table extraction (TE), which allows for understanding linear and formatted data. One exception is the pdftotext control, which solely used the embedded text layer artifact and did not have any OCR stage.

1) Multi-model pipelines: **Marker** [52] (Version 1.0) is a multi-model pipeline from Datalab, which uses Surya [53] for OCR, and its own tool for table extraction, trained on FinTabNet [54]. **Docling** [55] (2.0) by IBM, is another multi-model pipeline, which uses DocLayNet and TableFormer [56], [57]. It was chosen to pair with Marker to minimize the effect of choice-of-pipeline.

2) VLMs: **Qwen2.5-VL** [16], [58] is a VLM in the Qwen-VL [58] family, which was the first to integrate a native dynamic-resolution Vision Transformer, allowing better TE. Because it is a VLM, the output may be generalized based on the training data; for instance, it could suggest species names that are similar to its training, instead of staying entirely aligned with the input. **Qwen3-VL** [17], [58] uses DeepStack [59] to interleave tokens from the visual layer to their aligned transformer layers, improving output generally. This model also does better with poor resolution or misaligned text. **Qwen3.5-9B** [60] is a newer generation model in the Qwen [58] family. We were unable to run Qwen3.6 [61], [62] on our GPU grid (see details below in Section III-D). **olmOCR** [63] (2-7B-1025) is a VLM that has been fine-tuned on Qwen-2.5-VL-7B, but incorporates reinforcement learning with verifiable rewards, improving performance on tables. Its inclusion isolates the effect of domain-specific fine-tuning. **nougat** [64] (v: base) is a VLM based on Donut [18] (used by [10], as mentioned) which lacks a separate OCR stage, as the transformer decoder integrates sooner in the pipeline directly on the image. Nougat is trained on academic documents, and may hallucinate more for novel inputs. It is included here as a more specialized version of the general-document VLMs like Qwen3. **Pix2Struct** [65] (v: google/pix2struct-large) is another VLM, but it lacks the chat-style interface of the generalist models, and is specifically trained on web-pages to decode HTML structure and content. Because it is smaller and finely-tuned, it is included to anchor the possibility that the finely-trained VLMs outperform generalist models, compared to the traditional rule-based baselines.

3) Multi-model VLM hybrid: **Chandra** [66] is the updated version of Marker [52], built on Qwen3.5-9B [60]. It comes with a native, immutable prompt to extract tables into HTML. We automatically convert these to CSV to enable scoring.

4) Rule-based, deterministic tools: **Tesseract** [67]–[71] (v5.3) is included as a traditional OCR engine operating directly on the page raster. **pdfplumber** [47] (v0.11) uses rule-based heuristics. It struggles with complex tables, and, like Camelot, it depends upon the extracted text-layer embedded in a PDF by text-extractors [72]. **Camelot** [73] (v0.11) also depends upon the PDF text-layer, and is used as an alternative to pdfplumber, and is more finely-tuned for TE.

5) Text-dump control: **pdftotext** [74] was used as a control: it only reads the text layer embedded in the PDF by Adobe

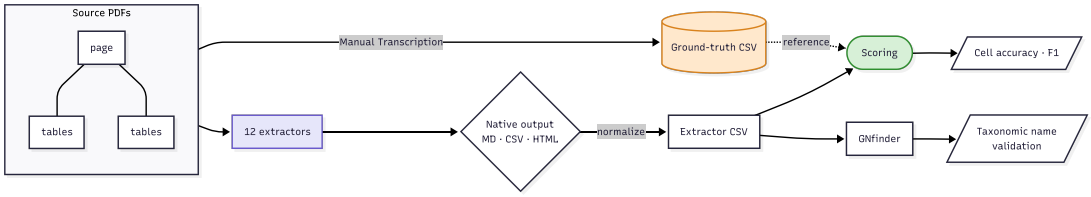


Fig. 5. The main PDF extractor pipeline.

Acrobat 6 [48] when the initial scans were made. There is no OCR, TE, or VLM.

To isolate tool errors, we evaluate every extractor through a common evaluation harness — a uniform runner, per-tool adapters, and a shared scoring pipeline. Each run was idempotent and logged. Empty results were logged as warnings, to avoid conflation with failed searches. Invocations were limited to a 1-hour timeout. All outputs were written atomically to a temp file, and renamed and moved on completion. Each extractor has a `manifest.json` which records checksums per output, as well as the log. For each VLM, we log the HuggingFace [75] commit hash at load time, for reproducibility and provenance. Tool configurations were centralized in a single config.

Each extractor ran in its own isolated environment. The evaluation harness also normalized for rotation errors. One of the difficulties of the harness involved cache integrity from HuggingFace, as caches needed to be redownloaded for each server, and caches could end up corrupted on different CUDA instances, resulting in run failures. Finally, the VLM tools (Qwen* families, olmOCR) received an extraction prompt, while Chandra already has a baked-in prompt[1].

D. Full corpus run

After the comparative evaluation, Chandra, Marker, Qwen3-VL, and olmOCR extractors are run on the full corpus. Per-page sharding of 36 PDFs (404 pages) ensured balance across GPUs for large runs. The ECS GPU grid at Te Herenga Waka Victoria University of Wellington contains 30 servers with 80 GPUs, running CUDA [76]. The most common GPUs are NVIDIA RTX A5000s (24GB); some runs were also on an NVIDIA A40 with 46GB of GPU memory.

E. Species verification

The tables hold three types of information: headers and captions, values, and scientific names. Names which appear in a table should also reliably appear in the paper main text; names that do not are likely extraction errors. To catch these, we use GNfinder [77], [78], the state-of-the-art named-entity-recognition software for scientific names. We run GNfinder on two per-page bundles: the pooled table extractions from Chandra and Qwen3-VL and the GT, and on olmOCR's extractions of the running text as an independent pipeline not from Datalab.

[1]Prompts are visible at https://anonymous.4open.science/r/bps-anonymous-data-D881.

A table name found in no paper's corroborating source is an *uncorroborated name* (UCN). Names with <5 hits (near-UCN) are possible extraction failures, and flagged for visual review. Names are matched based on token-prefix matching: genera and species-names must agree, but genus-only names are possible. Abbreviated generic names are expanded. This avoids UCNs caused by formatting differences. The parser uses regex, which is brittle at scale; at least five regex bugs were caught by a post-hoc test suite.

We also run another check, by comparing names against all GNfinder-extracted names from the NZ checklist [40] (5553 names, including historical synonyms).

The audit was iterated through four steps: (1), all three tools were pooled, which caused 0 UCNs due to the full text being a superset of the table set; (2), only a Marker-run corroborating source was used, causing 41 UCNs, which were mostly false positives; (3) bidirectional token-prefix matching was added, reflecting 5 defensible UCNs; (4) olmOCR was used as the control, along with the <5 threshold.

F. Scoring

Tool scores are compared against the GT with a bias-corrected and accelerated (BCa) bootstrap (10,000 resamples, $\alpha = 0.05$, seed 42) on per-table cell accuracy; the 300→400-dpi change in the final column of Table I uses a paired BCa bootstrap over the same tables. Tools scoring categorical zero across all 21 tables are reported with the rule-of-three upper bound (0, 0.143) rather than a degenerate [0, 0] interval.

IV. Results

A. Comparative evaluation results

The best-performing extractor in terms of cell accuracy was Chandra, as shown in Table I. Recall separates recovered content from a reconstructed table, and recall results were split. Qwen3.5 was the best performer for row recall by a small margin (while being significantly lower in cell accuracy); olmOCR was the best for column recall.

Chandra, olmOCR, Marker, and Qwen3-VL all produced output for 404/404 pages. However, as shown in Table II, success on reading a page does not equate to success on extracting a table. Both Qwen3-VL and Marker detect almost the exact amount of tables; Chandra, however, extracts an extra 22 tables. These were line-chart figures which Chandra converted to HTML tables. No other model extracts values from these tables; Chandra moves the goal-post for extraction success, rather than over-extracting unrelated content. Chandra

TABLE I

COMPARATIVE EVALUATION RESULTS ON GT TABLES ($n = 21$). EACH METRIC CELL: MACRO MEAN [95% BCA CI] / CELL-WEIGHTED MEAN. PDFTOTEXT IS A RAW TEXT-DUMP CONTROL. FINAL COLUMN: CHANGE IN MACRO CELL-ACCURACY FROM 300→400-DPI RESCAN (PAIRED BCA); * = 95% CI EXCLUDES 0. 400-DPI CONTRASTS NOT ADJUSTED FOR MULTIPLICITY; TREAT * AS SUGGESTIVE. TEXT-LAYER TOOLS (CAMELOT, PDFPLUMBER, PDFTOTEXT) EXCLUDED; 400-DPI SCANS HAVE NO OCR LAYER.

Extractor	Cell accuracy	Row recall	Column recall	ΔCell acc. at 400dpi
Chandra (chandra-ocr-2)	**0.881** [0.78, 0.95] / 0.871	0.861 [0.71, 0.92] / 0.849	0.615 [0.45, 0.77] / 0.573	-0.098 [-0.25, +0.01]
Marker 1.0	0.535 [0.36, 0.69] / 0.544	0.829 [0.70, 0.91] / 0.818	0.562 [0.38, 0.72] / 0.584	-0.155 [-0.35, -0.00]*
olmOCR-2-7B	0.534 [0.36, 0.70] / 0.492	0.781 [0.55, 0.92] / 0.733	**0.721** [0.51, 0.86] / 0.673	+0.155 [+0.05, +0.32]*
Qwen3-VL-8B	0.492 [0.35, 0.63] / 0.461	0.704 [0.52, 0.83] / 0.625	0.712 [0.54, 0.84] / 0.701	+0.110 [-0.08, +0.30]
Qwen3.5-9B	0.346 [0.19, 0.53] / 0.324	**0.869** [0.76, 0.93] / 0.836	0.444 [0.25, 0.64] / 0.400	+0.176 [-0.03, +0.39]
Docling 2.0	0.276 [0.15, 0.46] / 0.219	0.561 [0.39, 0.70] / 0.507	0.405 [0.23, 0.60] / 0.386	+0.130 [-0.09, +0.31]
Qwen2.5-VL-7B	0.146 [0.05, 0.33] / 0.081	0.183 [0.05, 0.38] / 0.132	0.133 [0.04, 0.31] / 0.074	+0.136 [-0.03, +0.31]
Camelot 0.11	0.054 [0.00, 0.26] / 0.016	0.200 [0.09, 0.36] / 0.167	0.017 [0.00, 0.06] / 0.006	— *(text-layer)*
nougat-base	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
pdfplumber 0.11	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	— *(text-layer)*
Pix2Struct-large	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
Tesseract 5.3	0.000 [0.00, 0.14] / 0.000	0.383 [0.23, 0.54] / 0.394	0.000 [0.00, 0.14] / 0.000	+0.000 [+0.00, +0.00]
pdftotext (poppler) *(control)*	0.000 [0.00, 0.14] / 0.000	0.309 [0.16, 0.49] / 0.290	0.000 [0.00, 0.14] / 0.000	— *(text-layer)*

also alone extracted a two-column list of species from [79] as a table.

The PDFs contained a text layer from Adobe Acrobat 6 [48], which is included in Table I through the use of pdftotext, a dumb extractor that does no OCR and only reads the embedded text. The content is structureless, so cell accuracy is null. For extractors that do not use this layer, the recall performance gains by Chandra and Marker show above a 50% increase in general data rescue. Chandra is subsequently used as the extractor of the full text of the papers.

Pdfplumber produced no table output for any table. Nougat produced 21 tables, but 13 were empty or contained hallucinated text, and there was no recall. Pix2Struct likewise produced 2 tables, with no recall. Camelot recovered a few correct cells, despite no raster layout clues in scanned images, but was likewise below 20% across the board, as it could not determine structure. For instance, in 1997T4, it scored 0.9706 for row recall, but 0.0 for column recall, as it could not interpret whitespace appropriately on an ungridded table.

Chandra outputs HTML; Marker and Docling output markdown, and the rest CSV. We reran Marker and Docling through Chandra's HTML → CSV path, and found mild cell accuracy bumps: +0.044 (Marker 0.535 → 0.579), -0.034 (Docling 0.276 → 0.242). The ∼30-point gap for Chandra versus the mid-range extractors is due to model quality, not output representation. Likewise, some tools require preprocessing handling, such as rotated scans, while others can autorotate their inputs; here, the harness covered these issues, and they were not considered as eliminating factors.

Within-family differences were occasionally important. Qwen2.5 collapsed A-schema tables which had kilometers and bird numbers paired in two rows, while Qwen3 kept them separate, improving recall and accuracy. However, Marker and Qwen3-VL were indistinguishable across the three metrics. The training set also matters: nougat consistently hallucinated with mathematical, LaTeX-flavored code, as it was trained on arXiv and PMC documents [64].

The 400dpi scan resulted in few substantive differences overall, as shown in the last column of Table I.

Chandra, Marker, olmOCR, and Qwen3-VL ran at comparable per-GPU throughput (∼1.3 pages/min). With 4-way page sharding the corpus completed in ∼1.3h of active compute on an NVIDIA A40; Chandra was run as a single stream (∼5h) on an RTX A6000. These numbers are reconstructed from per-page completion timestamps; they may also fail to reflect load and restart times from configuration attempts. Scoring and GNfinder evaluation times were trivial even on CPUs.

B. Errors in the GT tables

There are 21 GT tables from six years (1960, 1963, 1974, 1984, 1990, and 1997); the entire corpus from 1960-1999 has 36 papers with 404 pages. The GT contains 10,321 GT internal cells and 2,920 non-empty data values, and contains 118 unique names (104 matched to the BirdsNZ Checklist [40]). The processing and extracting of the GT tables highlighted 21 different errors in 12/21 (57%) D2 tables (0.19% of the total 10,982 GT cells including headers). These errors include (with the amount in brackets):

- **Numerical typos** [7]. For instance, 1984 table 1 (1984T1) printed '29', but should have been '20'.
- **Text typos** [2]. In 1974T2, 'WDI' should be 'WD' (for Westland).
- **Incorrect names** [2]. In 1997T2, *Fregatta tropica* should be *Fregetta*; this inaccurate spelling has been published elsewhere [80]. Spelling errors have a particular status under the International Code of Zoological Nomenclature [81] governing scientific names; for example, all known spelling errors for names are logged in the BirdsNZ checklist [40] (see Section IV-C). *Anous tenuirostris minutes* (not *minutus*) appears new to science.
- **Row shifts** [3]. In 1963T3, the totals for *P. turtur*, *Pachyptila. sp.**, and *Puffinus carneipes* were swapped.
- **Simple arithmetic errors** [2]. In 1984T3, the total was 14065, when it ought to have been 14066.
- **Cascading arithmetic errors** [2]. In 1974T3, one row had +1, while the total had -1, for a net zero.
- **Missing content** [2]. In 1984T2, the months row for *Oceanites* should include '(2)' for May, but doesn't.

TABLE II

CORPUS TABLE DETECTION VS. A MANUAL PAGE-LEVEL CHECK ($N = 404$ PAGES, 36 PAPERS). THE CHECK FINDS 179 PAGES WITH A TABLE (44.3%) AND 25 PAGES WITH A LINE-GRAPH. Δ IS EACH TOOL'S DETECTED-TABLE COUNT MINUS THE MANUAL TABLE COUNT; CHANDRA'S POSITIVE Δ INDICATES EXTRACTION OF LINE-GRAPH FIGURES AS TABLES.

Tool	Tables detected	Δ vs check	What it captures
Manual scan (GT_2)	179 (25)	—	manual page-level counts (# of line graphs in parens)
Chandra	202	+23	tables + line-graph figures→data
Qwen3-VL	178	-1	tables
Marker	176	-3	tables; some narrative on non-table pages
olmOCR	154	-25	tables (under-detects)

Likewise, in 1997T2, *P.melanoleucos* is missing a space between the generic name and the specific name.

- **Data type errors** [1]. In 1997T4, 'i6' should be '16'.
- **Printing artifacts**. Printing may also cause errors; for instance, in 1965T2, *Larus bulleri* is almost unreadable, as the *r* is essentially erased by poor copying from a typewriter. The name is readable from context by a human, but may present uncountable difficulties for a naïve extractor.

No single error dominated the GT, and different tools may pick up some errors better (e.g., species-name vs. arithmetic errors). Some errors cannot be picked up by cross-table checks; e.g., in 1984, totals reconciled across tables, but each contained row/column errors that were only picked up by internal checks.

C. Species verification result

GNfinder was run on all of the GT tables by Chandra and Qwen3-VL. Any names found in the tables, but not found in the output of another extractor (the corroborator text) run on the full text of all 36 pages, are considered uncorroborated names (UCNs). For a run with Marker as the corroborator, 391 names were found, with 124 uncorroborated, 5 of which are valid names checked against the BirdsNZ Checklist; for a run with olmOCR, 358 unique names were found in the tables, with 144 UCNs, 23 of which are valid. There was a four-name intersection between these two valid subsets: *Fulmarus glacialis*, *Gavia*, *Sternula*, and *Puffinus gavia gavia*. The last is an uninteresting result, as an old subspecies that is no longer recognized; the single-names are likewise predictable, as rare genera. *Gavia* is a misparsing by GNfinder, as it is a name for a northern hemisphere genus (loons) that is shared with a species name for Fluttering Shearwater *Puffinus gavia*, a local species in NZ.

Fulmarus glacialis is another type of error. The Southern Fulmar *Fulmarus glacialoides* is related to the Northern Fulmar *Fulmarus glacialis*. Most VLMs consistently output the northern species instead of the southern one when reading the tables: Qwen3-VL output *glacialoides* (39 times) correctly, but *glacialis* (7) and *glacialisoides* (4); Chandra *glacialis* (44) or *glacialisoides* (9); olmOCR *glacialisoides* (62) consistently. Only Marker, which lacks a VLM layer interlocked with the OCR extraction, consistently output the correct *glacialoides* (60) every time.

The first extralimital record of a Northern Fulmar in NZ was in 2014 [40]; there should be no references to *glacialis* in any of these texts. This is almost certainly a training error, where more texts from the northern hemisphere involving a more studied species were in the training data, and the VLMs consistently either truncated the species name to match, or attempted a reading like the incorrect *glacialisoides*. The olmOCR-sourced 23 valid names flagged as UCNs compared to Marker (5) is also likely a result of VLM-introduced noise - the main example again being that olmOCR failed to read *glacialoides* correctly even once, flagging it as a UCN. Cross-tool audits have to take into account tool differences for internal checks.

Many OCR-garbled names were caught during this check: for instance, *Daotion capensis* for Cape Petrel *Daption capensis*. These fail a cross-check to BirdsNZ, while *Fulmarus glacialis* does appear in [40] as a valid name and would be missed by a single authoritative check. Multiple checks provide multiple safeguards, as in Section IV-B where arithmetic checks isolated errors in row/column sums. There are no possible checks on the cell numbers as individual records, except to refer to the digitized D1, which is not always in agreement with D2.

V. CONCLUSIONS

Initially, an attempt was made to transcribe all tables by hand, including using paid subcontractors. Each table took around 30 minutes to hand-verify, and manual processing was error-prone, due to difficulty with scientific names and human task fatigue. The switch to local VLMs was an obvious choice, but a lack of clear off-the-shelf winners of comparative evaluations on heterogeneous scientific tables motivated this study.

The comparative evaluation, where Chandra is the clear winner for GPU-able runs, is only part of a broader pipeline. There is no single extractor which can be used on this data; the highest gains in verifiability came from using multiple extractors from different families to validate textual errors. Choosing a single tool is brittle; choosing several tools and building a harness for their evaluation allows consistent feedback loops that clean the data, and which point out errors that are not caught by manual validation, such as the *i6* for '16' error that neither the printer nor we caught before validators did. More operational recommendations include detecting DPI and text layers before runs; preserving outputs at each step of the process; treating the harness and testing as first-class citizens during an evaluation pipeline; and engaging with domain experts to verify content that is difficult to verify

through external sources (in this case, *Fulmarus glacialis* and the BirdsNZ checklist).

There are limitations to this study. The GT (n=21) is relatively small; the paper uses a single corpus; and only one author served as an annotator. Further, the outputs could not be released at this time due to legal considerations. The initial GT provenance was mixed between VLM and manual usage, which may have introduced errors or bias. There is also a structural issue where the gap between D1 and D2 is impossible to bridge without reference to the original index cards. Other points exist: largely single-family VLM usage, but not the cutting edge Qwen3.6; a single raster typeset datapoint (1997-9); and the operational difficulty of sharding and accessing CUDA instances on the local grid.

A. Future Work

Future work would include cleaning the full corpus, removing overlapping logs in D1 and comparing the values in D2 against D1, and producing homogeneous output formats. More future work includes creating a historic API of names; e.g., the White-flippered Little Penguin *Eudyptula minor albosignata* was a near-uncorroborated name but the name is now no longer in use, as the subspecies is not recognized by NZ authorities [40]. Some work in this field, such as named entity recognition for scientific names [82], could be applied here. It may be possible to compare trends across years, which was not done consistently in the original summaries, although later papers highlighted long-term differences for some genera, as in [83], [84] for *Pterodroma* and *Pachyptila*.

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Claude.ai [25] was used in the development of the code base. AI output was not used in the writing of the paper, but AI was used to scaffold out Figure 5 in Mermaid, and in writing Python scripts to create formatted LaTeX tables. Claude was instructed to make suggestions as bullet points, but not to provide wording or content; this prompt was repeated when it erred. No Claude access was given to Overleaf [85], and AI features in Overleaf and Google and Google Sheets were ignored. No AI bibtex references were used, and all were manually checked; scripts were used to verify all DOIs against Crossref.

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